

Data-Centric Evolution in Autonomous Driving: A Comprehensive Survey of Big Data System, Data Mining, and Closed-Loop Technologies

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Abstract

The aspiration of the next generation’s autonomous driving (AD) technology relies on the dedicated integration and interaction among intelligent perception, prediction, planning, and low-level control. There has been a huge bottleneck regarding the upper bound of autonomous driving algorithm performance, a consensus from academia and industry believes that the key to surmount the bottleneck lies in data-centric autonomous driving technology. Recent advancement in AD simulation, closed-loop model training, and AD big data engine have gained some valuable experience. However, there is a lack of systematic knowledge and deep understanding regarding how to build efficient data-centric AD technology for AD algorithm self-evolution and better AD big data accumulation. To fill in the identified research gaps, this article will closely focus on reviewing the state-of-the-art data-driven autonomous driving technologies, with an emphasis on the comprehensive taxonomy of autonomous driving datasets characterized by milestone generations, key features, data acquisition settings, etc. Furthermore, we provide a systematic review of the existing benchmark closed-loop AD big data pipelines from the industrial frontier, including the procedure of closed-loop frameworks, key technologies, and empirical studies. Finally, the future directions, potential applications, limitations and concerns are discussed to arouse efforts from both academia and industry for promoting the further development of autonomous driving. The project repository is available at: <https://github.com/LincanLi98/Awesome-Data-Centric-Autonomous-Driving>.

1 Background Introduction and Motivation

The advancement of deep learning technology combined with hardware supercomputing capacity and big data collection ability, has aroused the emergence of autonomous driving algorithms, which involve a range of task modules such as detection [Wu *et al.*, 2023b; Hou *et al.*, 2023], tracking [Zhang *et al.*, 2023b; Li and Jin, 2022], occupancy forecasting [Khurana *et al.*, 2023; Zheng *et al.*, 2023], trajectory prediction [Wang *et al.*, 2022; Zhang *et al.*, 2023a], motion planning [Huang *et al.*, 2023; Chen *et al.*, 2023a], and low-level control [Kim *et al.*, 2022]. To achieve the ultimate goal of safe, adaptive, and efficient autonomous driving, all of the algorithms and softwares must be built upon a highly automated closed-loop data driven pipeline [Zhang *et al.*, 2022a; Chen *et al.*, 2023b] for self-evolution and upgrading. This is the pre-requisite of the predicted large-scale commercial application of autonomous vehicle (AV) technology in year 2025 [Khan *et al.*, 2022]. Despite extensive research works have done to solely focus on designing AD algorithm itself using either open-source datasets or self-collected datasets, little efforts have been devoted to developing a systematic technology pipeline of data-centric autonomous driving, which in turn give negative impacts to the entire systems’ performance improvement, resulting in the existing bottleneck of AD algorithm performance upper-bound. The above mentioned concerns motivate us to carry out a comprehensive review regarding the state-of-the-art data-centric autonomous driving technologies, including AD big data system, AD data mining, and closed-loop AD technologies. By adopting this research, we anticipate that it will enhance the industrial R&D process and algorithm performance of autonomous driving, while also providing valuable insights to related academic studies in the same field.

Data is the new oil in autonomous driving world. There has been a severe Long-Tail Distribution problem in real-world autonomous driving applications [Long *et al.*, 2022]. In order to fully address the long-tail distribution problem, it is